1

Introduction

1.1 Historical Parentage

There are many opinions about the origin of chemometrics. Until quite recently, the birth of chemometrics was considered to have happened in the 1970s. Its name first appeared in 1972 in an article by Svante Wold [1]: in fact, the topic of this article was not one that we would recognise as being core to chemometrics, being relevant to neither multivariate analysis nor experimental design. For over a decade, the word chemometrics was considered to be of very low profile, and it developed a recognisable presence only in the 1980s, as described below.

However, if an explorer describes a new species in a forest, the species was there long before the explorer. Thus, the naming of the discipline just recognises that it had reached some level of visibility and maturity. As people re-evaluate the origins of chemometrics, the birth can be traced many years back.

Chemometrics burst into the world due to three fundamental factors, applied statistics (multivariate and experimental design), statistics in analytical and physical chemistry, and scientific computing.

1.1.1 Applied Statistics

The ideas of multivariate statistics have been around a long time. R.A. Fisher and colleagues working in Rothamsted, UK, formalised many of our modern ideas while applying primarily to agriculture. In the UK, before the First World War, many of the upper classes owned extensive land and relied on their income from tenant farmers and agricultural labourers. After the First World War, the cost of labour became higher, with many moving to the cities, and there was stronger competition of food from global imports. This meant that historic agricultural practices were seen to be inefficient and it was hard for landowners (or companies that took over large estates) to be economic and competitive, hence a huge emphasis on agricultural research, including statistics to improve these. R.A. Fisher and co-workers published some of the first major books and papers that we would regard as defining modern statistical thinking [2, 3], introducing ideas ranging from the null hypothesis to discriminant analysis to ANOVA. Some of the work of Fisher followed from the pioneering work of Karl Pearson in the University College London who had founded the world's first statistics department previously and had first formulated ideas such as p values and correlation coefficients.

During the 1920s and 1930s, a number of important pioneers of multivariate statistics published their work, many strongly influenced or having worked with Fisher, including Harold Hotelling, credited by many as defining principal components analysis (PCA) [4], although Pearson had independently described this method some 30 years ago, but under a different guise. As so often ideas are reported several times over in science, it is the person that names it and popularises it that often gets the credit: in the early twentieth century, libraries were often localised and there were very few international journals (Hotelling working mainly in the US) and certainly no internet; therefore, parallel work was often reported.

The principles of statistical experimental design were also formulated at around this period. There had been early reports on what we regard as modern approaches to formal designs before that, for example James Lind's work on scurvy in the eighteenth century and Charles Pierce's discussion on randomised trials in the nineteenth century, but Fisher's classic work of the 1930s put all the concepts together in a rigorous statistical format [5].

Much non-Bayesian, applied statistical thinking has been based on principles established in the 1920s and 1930s, for nearly a century. Early applications include agriculture, psychology, finance and genetics. After the Second World War, the chemical industry took an interest. In the 1920s, an important need was to improve agricultural practice, but by the 1950s, a major need was to improve processes in manufacturing, especially chemical engineering; hence, many more statisticians were employed within the industry. O.L. Davies edited an important book on experimental design with

Chemometrics: Data Driven Extraction for Science, Second Edition. Richard G. Brereton.
© 2018 John Wiley & Sons Ltd. Published 2018 by John Wiley & Sons Ltd.
Companion website: http://booksupport.wiley.com

contributions from colleagues in ICI [6]. Foremost was G.E.P. Box, son-in-law of Fisher, whose book with colleagues is one of the most important post-war classics in experimental design and multi-linear regression [7].

These statistical building blocks were already mature by the time people started calling themselves chemometricians and have changed only a little during the intervening period.

1.1.2 Statistics in Analytical and Physical Chemistry

Statistical methods, for example, to estimate accuracy and precision of measurements or to determine a best-fit linear relationship between two variables, have been available to analytical and physical chemists for over a century. Almost every general analytical textbook includes chapters on univariate statistics and has done for decades. Although theoretically we could view this as applied statistics, on the whole, the people who advanced statistics in analytical chemistry did not class themselves as applied statisticians and specialist terminology has developed over time.

Most quantitative analytical and physical chemistry until the 1970s was viewed as a univariate field; that is, only one independent variable was measured in an experiment. Usually, all other external factors were kept constant. This approach worked well in mechanics or fundamental physics, the so-called 'One Factor at a Time' (OFAT) approach. Hence, statistical methods were primarily used for univariate analysis of data. By the late 1940s, some analytical chemists were aware of ANOVA, *F*-tests and linear regression [8], although the term *chemometrics* had not been invented, but multivariate data came along much later.

There would have been very limited cross-fertilisation between applied statisticians, working in mathematics departments, and analytical chemists in chemistry departments, during these early days. Different departments often had different buildings, different libraries and different textbooks. A chemist, however numerate, would feel a stranger walking into a maths building and would probably cocoon him or herself in their own library. There was no such thing as the Internet or Web or Knowledge or electronic journals. Maths journals published papers for mathematicians and vice versa for chemistry journals. Although in areas such as agriculture and psychology there was a tradition of consulting statisticians, chemists were numerate and tended to talk to each other – an experimental chemist wanting to fit a straight line would talk to a physical chemist in the tea room if need be. Hence, ideas did not travel in academia. Industry was somewhat more pragmatic, but even there, the main statistical innovations were in chemical engineering and process chemistry and often classed as industrial chemistry. The top Universities often did not teach or research industrial chemistry, although they did teach Newtonian physics and relativity. In fact, the treatment of variables and errors by physicists trying, for example, to measure gravitational effects or the distance of a star is quite different to multivariate statistics: the former try to design experiments so that only one factor is studied and to make sure any errors are minimised and from one source, whereas a multivariate statistician might accept and expect data to be multifactorial.

Hence, statistics in analytical chemistry diverged from applied statistics for many decades. Caulcutt and Body's book first published in 1983 contains nothing on multivariate statistics [9] and in Miller and Miller's book of 1993 just one out of six main chapters is devoted to experimental design, optimisation and pattern recognition (including PCA) [10].

Even now, there are numerous useful books aimed at analytical and physical chemists that omit multivariate statistics. An elaborate vocabulary has developed for the needs of analytical chemists, with specialist concepts that are rarely encountered in other areas. Some analytical chemists in the 1960s to 1980s were aware that multivariate approaches existed and did venture into chemometrics, but good multivariate data were limited. Most are aware of ANOVA and experimental design. However, statistics for analytical chemistry tends to lead a separate existence from chemometrics, although multivariate methods derived from chemometrics do have a small foothold within most graduate-level courses and books in general analytical chemistry, and certainly quantitative analytical (and physical) chemistry was an important building block for modern chemometrics.

Over the last two decades, however, applications of chemometrics have moved far beyond traditional quantitative analytical chemistry, for example, into the areas of metabolomics, environment, cultural heritage or food, where the outcome is not necessarily to measure accurately the concentration of an analyte or how many compounds are in the spectra of a series of mixtures. This means that the aim of some chemometric analysis has changed. We often do not always have, for example, well-established reference samples and, in many cases, we cannot judge a method by how efficiently it predicts properties of these reference samples. We may not know whether the spectra of some extracts of urine samples can contain enough information to tell whether our donors are diseased or not: it may depend on how the disease has progressed, how good the diagnosis is, what the genetics of the donor and so on. Hence, we may never have a model that perfectly distinguishes two groups of samples. In classical physical or analytical chemistry, the answer is usually known to a greater accuracy than we can predict, in advance, so we can always tell which methods

are best. This gradual change in culture distinguishes much of modern chemometrics from traditional statistics in analytical chemistry, although analytical chemistry is definitely one of the ancestors of chemometrics, and the two are symbiotic.

1.1.3 Scientific Computing

Another revolution happened as from the 1960s, the use of computers in scientific research.

Many of the original statistical computations required complex matrix operations that may have taken days or even weeks to solve using manual calculations even with calculators or slide rules. This limited the applicability of many statistical methods. Many early statistical papers were intensely theoretical and some methods were applied only to important and economically significant problems: an agricultural experiment that took several years deserved a couple of weeks manually computing the trends in the data. However, they were not widespread, especially in scientific laboratories.

With the 1960s, scientists in the best resourced laboratories gained access to mainframe computers. Usually, they had to be programmed in languages such as FORTRAN and used punch cards, paper tape and line printers. However, they allowed a rapid adoption of computers by applied scientists, which became the third revolution that led to chemometrics.

Resolution and rank analysis of spectroscopy of mixtures had its vintage in the 1960s, with a small number of pioneering papers [11, 12] taking advantage of newly available computer power: in earlier papers, such methods were reported but were applied to very small problems, for example, of four mixtures and four wavelengths due to the difficulty of manual calculation. Multivariate spectroscopic resolution developed quite separately to multivariate statistics, primarily via physical chemistry. The original terminology differed quite considerably from statistics and was primarily that of physics.

Over the 1960s and 1970s, there were many papers about spectroscopic resolution in both the physical chemistry and the analytical chemistry literature, but Ed Malinowski, whose remarkable publication career stretches from 1955 to 2011, is best recognised to having put these concepts together with multivariate statistics. He published what many regard as the first book that covered one important area of chemometrics [13], which he called *Factor Analysis*, involving determining the number of components in spectroscopic mixtures together with their characteristics.

Meanwhile, a separate development in scientific computing emerged in the 1960s – partly catalysed by NASA's trip to the moon – to use AI to identify compounds spectroscopically [14], a project that involved Nobel Prize winners and spawned the whole new area of expert systems. This in turn led to the field of pattern recognition and the award of several competitive grants in scientific computing, particularly in the USA.

Isenhour, Jurs and Kowalski were early pioneers of computerised learning in chemistry, primarily using pattern recognition [15], the early group founded by Isenhour, who left in 1969. Kowalski took over the reins in 1974, initially with an interest in chemical pattern recognition.

Hence, computational chemistry arrived via both physical chemistry of spectroscopic mixtures and organic chemistry for pattern recognition and had important elements in the formative mix in the 1960s and 1970s. This allowed the application to comparatively large problems and wider access to algorithms that had previously been rather theoretical.

1.2 Developments since the 1970s

Chemometrics slowly gained an identity from the mid-1970s, after Wold first named it. However, some of the recognised pioneers were slow to identify with it. For example, both Wold and Kowalski published far more papers using 'chemical pattern recognition' than 'chemometrics' as a keyword in the 1970s.

The first symposia with chemometrics in the name, in the USA, took place in the late 1970s. The first analytical chemistry review entitled 'Chemometrics' was published in 1980 [16]. The International Chemometrics Society was founded by Wold and Kowalski in the 1970s.

By this stage, although still relatively few workers identified themselves with chemometrics, small groups of enthusiasts were promoting the name and idea. In those days, most of those that identified themselves as chemometricians were quite expert programmers who cut their teeth on a mainframe or, latterly, primitive micros. Some even started their scientific careers before scientists had ready access to computers and may have had to learn programming via Assembly language so were in practice extremely good programmers. If a method was reported in a paper, the authors would typically program it in itself rather than using a package.

A NATO sponsored workshop in Cosenza, Italy, in 1983, brought together many of the early experts of the time [17] and events moved fast after that. The first journals dedicated to chemometrics, *Chemometrics and Intelligent Laboratory Systems* (Elsevier) and *Journal of Chemometrics* (Wiley), were founded in 1986 and 1987. Kowalski and co-workers produced the first comprehensive book in 1986 [18] followed by Massart and co-workers in 1988 [19]. Software packages such as Arthur, Unscrambler and Simca emerged during this period.

By the 1990s, well-established books, journals, courses and software were available, although still only quite a small number of dedicated groups worldwide. However, this changed when laboratory-based data started to become more readily available – the size of data sets and complexity of problems increased massively. In the 1980s, the emphasis was primarily on small problems such as the resolution of a cluster of HPLC peaks or deconvolution of uv/vis spectra. Economically important problems in process control and NIR spectroscopy posed new challenges to chemometricians and gradually moved the subject from a rather theoretical application of quantitative analytical chemistry to a more applied subject. There was a special interest in the interface between chemical engineering and chemometrics.

A further revolution has happened in the last 15 years when complex real-world data have become available. This has allowed looking at applications ranging from metabolomics to heritage studies to forensics and so on where large data sets are available. It has resulted in chemometrics tools becoming very widely used, although the core community of experts is probably no bigger than a few decades ago. The widespread applicability of common chemometric methods such as PCA, classification, calibration, and so on, leads to an urgent need to understand these methods. This book is primarily aimed at potential users who want to understand the underlying mathematical approaches, rather than just use packages.

1.3 Software and Calculations

The key to chemometrics is to understand how to perform meaningful calculations on data. In most cases, these calculations are too complex to do by hand or using a calculator; hence, it is necessary to use some software.

The approach taken in this book, which differs from many books on chemometrics, is to understand the methods using numeric examples. Some excellent books and reviews are more descriptive, listing the methods available together with the literature references and possibly some examples. Others have a big emphasis on equations and output from packages. This book, however, is primarily based on how I personally learn and understand new methods, and how I have found it most effective to help students working with me. Data analysis is not really a knowledge-based subject but is more a skill-based subject. A good organic chemist may have encyclopaedic knowledge of reactions in their own area. The best supervisor will be able to list to his or her students thousands of reactions, or papers, or conditions that will aid their students, and with experience this knowledge base grows. In chemometrics, although there are quite a number of named methods, the key is not to learn hundreds of equations but to understand a few basic principles. These ideas, such as multiple linear regression (MLR), occur again and again but in different contexts. To become skilled in chemometric data analysis, practice to manipulating numbers is required, not an enormous knowledge base. Although equations are necessary for the formal description of methods, and cannot easily be avoided, it is easiest to understand the methods in this book by looking at numbers. Hence, the methods described in this book are illustrated using numerical examples, which are available for the reader to reproduce. The data sets employed in this book are available on the publisher's website. In addition to the main book, there are extensive problems at the end of each main chapter. All numerical examples are quite small and are designed in such a manner that you can check all the numbers yourselves. Some are reduced versions of larger data sets, such as spectra recorded at 5 nm rather than 1 nm intervals. Many real examples, especially in chromatography and spectroscopy, simply differ in size to those in this book. In addition, the examples are chosen so that they are feasible to analyse fairly simply.

One of the difficulties is to decide the software to be employed in order to analyse the data. This book is not restrictive and you can use any approach you like. Some readers may like to program their own methods, for example, in C or Visual Basic. Others may like to use statistical packages such as SAS or SPSS. There is a significant statistical community that uses R. Some groups use ready packaged chemometrics software such as Pirouette, Simca, Unscrambler, PLS Toolbox and several others in the market. One problem with using packages is that they are often very focussed in their facilities. What they do, they do excellently, but if they cannot do what you want, you may be stuck, even for relatively simple calculations. If you have an excellent multivariate package but want to use a Kalman filter, where do you turn? Perhaps you have the budget to buy another package, but if you just want to explore the method, the simplest implementation takes only an hour or less for an experienced Matlab programmer to implement. In addition, there

1.4 Further Reading

A large number of books and review articles have been written, covering differing aspects of chemometrics, often aimed at a variety of audiences. In Sections 1.1 and 1.2, we list some of the more historic books and papers. This chapter summarises some of the most widespread and recent works. In most cases, these works will allow the reader to delve further into the methods introduced within this book. In each category, only a few main books will be mentioned, but most have extensive bibliographies, allowing the reader to access information especially from the primary literature. Although there are also internet resources and numerous tutorial and review papers, in order to restrict the bibliography, we only list books.

1.4.1 General

Largest authored book in chemometrics is published by Massart and co-workers, part of two volumes [20, 21]. These volumes provide an in-depth summary of many modern chemometric methods, involving a wide range of techniques, and many references to the literature. The first volume, although, is quite strongly oriented towards analytical chemists but contains an excellent grounding in basic statistics for measurement science. The books are especially useful as springboards for the primary literature. This is a complete rewrite of the original book published in 1988 [19], which is still cited as a classic in the analytical chemistry literature. *Comprehensive Chemometrics* [22] is a follow-on from the same publisher, an encyclopaedic collection of edited articles in four volumes covering much of the knowledge base of chemometrics in 2009 and is probably the most comprehensive detailed summary of the subject.

Otto's book on chemometrics [23] is a well-regarded book now in its third edition covering quite a range of topics but at a fairly introductory level. The book looks at computing, in general, in analytical chemistry including databases and instrumental data acquisition. It is a very clearly written introduction for the analytical chemist, by an outstanding educator.

Beebe and co-workers at Dow Chemicals have produced a book [24] that is useful for many practitioners and contains very clear descriptions especially of multivariate calibration in spectroscopy and although some years old is still recommended for those working in this area. However, there is a strong 'American School' originating in part from the pioneering work of Kowalski in NIR spectroscopy and process control, and while covering the techniques required in this area in an outstanding way, and is well recommended as a next step for readers of this book working in this application area, it lacks a little in generality, probably because of the very close association between NIR and chemometrics in the minds of some.

Kramer has produced a somewhat more introductory book [25]. He is well known for his consultancy company and highly regarded courses, and his approach is less mathematical. This will suit some people very well, but may not be presented in a way that suits statisticians and chemical engineers.

This current author published a book on chemometrics at an early stage of the development of the subject [26], which has an emphasis on signal resolution and minimises matrix algebra, and is an introductory tutorial book especially for the laboratory-based chemist. This author also published a later book based on web articles that covers a range of applications as well as simple descriptions of methods [27]. This author has a series of ongoing short tutorial articles covering aspects of chemometrics as a column in *Journal of Chemometrics*, starting in 2014: these look more into the statistical principles of the subject.

The journal *Chemometrics and Intelligent Laboratory Systems* published regular tutorial review articles over its first decade or more of existence. Some of the earlier articles are good introductions to general subjects such as PCA, Fourier transforms and Matlab. They are collected together as two volumes [28, 29]. They also contain some valuable articles on expert systems.

Varmuza and Filtzmoser have published a very well-regarded book involving using R with clear descriptions within a statistical context [30]. Gemperline edited a multi-author book, which is currently in its second edition [31]. Pomerantsev has published a book oriented towards users of Excel [32]. Mark and Workman have written a comprehensive book aimed at spectroscopists [33]: it is very good at analytical instrumental chemistry and the authors are well regarded.

Meloun and Militky published a large book based on extensive course work [34]. This covers many topics in chemometrics and has a special feature of 1250 numerical problems and data sets.

Martens and Martens have produced a book that is quite a detailed discussion of how multivariate methods can be used in quality control [35] and covers several aspects of modern chemometrics, and so could be classed as a general book on chemometrics.

Although this list is not comprehensive, it lists most general books on chemometrics. There are also several books in different application areas such as food, environment, various types of spectroscopy and so on.

1.4.2 Specific Areas

There are a large number of books and review articles dealing with specific aspects of chemometrics, interesting as a next step after this book, and for a comprehensive chemometrics library. We will list just a few.

1.4.2.1 Experimental Design

In the area of experimental design, there are innumerable books, many written by statisticians. Specifically aimed at chemists, Deming and Morgan have produced a highly regarded book [36], which is well recommended as a next step after this book. Bayne and Rubin have written a clear and thorough book [37]. An introductory book mainly discussing factorial designs was written by Morgan as part of the Analytical Chemistry by Open Learning Series [38]. For mixture designs, involving compositional data, the classic statistical book by Cornell is much cited and recommended [39] but is quite mathematical. More historical books such as those by Fisher [5] and by Box and co-workers [7] have already been described above but are still relevant today.

1.4.2.2 Pattern Recognition and Principal Component Analysis

There are several books on pattern recognition and PCA. An introduction to several of the main techniques is provided in an edited book [40]. For more statistical in-depth descriptions of *Principal Components Analysis*, read books by Joliffe [41] and Mardia and co-authors [42]. An early but still valuable book by Massart and Kaufmann covers more than just its title 'cluster analysis' [43] and provides clear introductory material. Varmuza [44] and Strouf [45] wrote early books in the area when much of the rest of chemometrics was focussed on calibration and signal resolution.

A more up-to-date book focussed on pattern recognition was recently published by this author [46] that illustrated using several case studies. Over the past decade, there has been a much more interest in pattern recognition compared with a few decades ago, with increased application to areas such as metabolomics.

1.4.2.3 Multivariate Signal Analysis

Multivariate curve resolution (MCR) is the main topic of Malinowski's book [47], which is the third edition of his original book [13]. The author is a physical chemist and so the book is oriented towards that particular audience and especially relates to the spectroscopy of mixtures. Although there have been notable advances in the area, especially in alternating least squares (ALS), these are primarily published in the form of papers. Malinowski's book is still the classic book in the area. For more up-to-date reading, search for papers on MCR and ALS. However, the third edition of this book covers ALS well, and most of the pioneering papers were published some 15–20 years ago.

1.4.2.4 Multivariate Calibration

Multivariate calibration is a very popular area, and the much reprinted classic by Martens and Næs [48] is one of the most cited books in chemometrics. Much of the book is based around NIR spectroscopy, which was one of the major success stories in applied chemometrics in the 1980s and 1990s, but the clear mathematical descriptions of algorithms are particularly useful for a wider audience. The book by Beebe and co-workers [24] also has good in-depth discussion about calibration. A more recent book by Naes *et al.* is somewhat less theoretical and is mainly about multivariate calibration [49].

1.4.2.5 Statistical Methods

There are a number of books on general statistical methods in chemistry, mainly oriented towards analytical and physical chemists. Miller and Miller's book [10] has gone through several editions and takes the reader through many of the basic significance tests, distributions and so on. There is a small amount on chemometrics in the final chapter. The Royal Society of Chemistry published quite a nice introductory tutorial book by Gardiner [50]. Caulcutt and Boddy's book [9] is also a much reprinted and useful reference. There are several other competing books, most of which are very thorough, for example, in describing applications of the t-test, F-test and ANOVA but which do not progress much into modern chemometrics. If you are a physical chemist, Gans' viewpoint on deconvolution and curve fitting may suit you more [51], covering many regression methods. Meier and Zund published a book in 2000 [52] with a very thorough discussion of univariate methods especially in industrial practice and a little introduction to multivariate methods. Ellison and co-workers published a book based on the UK Valid Analytical Measurement initiative [53].

Several other books about statistical approaches (mainly univariate) in analytical chemistry and a number of international initiatives that regularly produce reports are regularly being developed.

1.4.2.6 Digital Signal Processing and Time Series

There are numerous books on digital signal processing (DSP) and Fourier transforms (FTs). Unfortunately, many of the chemically based books are fairly technical in nature and oriented towards specific techniques such as NMR; however, books written primarily by and for engineers and statisticians are often quite understandable. A recommended reference to DSP contains many of the main principles [54], but several similar books are available. A couple of recent general books on FTs are recommended [55, 56]. For non-linear deconvolution, Jansson's book is well known [57]. Methods for time series analysis are described in more depth in an outstanding and much reprinted book written by Chatfield [58].

1.4.2.7 Multi-way Methods

For chemometricians, the best book available is by Smilde *et al.* [59], which is a thorough description and illustration of the algorithms. There was much development in this area in the 1990s, which was a very exciting era for new algorithms, and the three authors were pioneers of some of the original papers in the chemometrics literature. This book is the best comprehensive summary of the application of such methods in chemistry.

References

1 Wold, S. (1972) Spline functions, a new tool in data-analysis. *Kemisk Tidskrift*, **3**, 34–37.
2 Fisher, R.A. (1925) *Statistical Methods for Research Workers*, Oliver and Boyd, Edinburgh.
3 Fisher, R.A. (1936) The use of multiple measurements in taxonomic problems. *Ann. Eugen.*, **7**, 179–188.
4 Hotelling, H. (1933) Analysis of a complex of statistical variables into principal components. *J. Educ. Psychol.*, **24**, 417–441.
5 Fisher, R.A. (1935) *The Design of Experiments*, Hafner, New York.
6 Davies, O.L. (ed.) (1956) *Statistical Methods in Research and Production*, Longman, London.
7 Box, G.E.P., Hunter, W.G. and Hunter, J.S. (1978) *Statistics for Experimenters*, John Wiley & Sons, Inc., New York.
8 Mandel, J. (1949) Statistical Methods in Analytical Chemistry. *J. Chem. Educ.*, **26**, 534–539.
9 Caulcutt, R. and Boddy, R. (1983) *Statistics for Analytical Chemists*, Chapman and Hall, London.
10 Miller, J.C. and Miller, J.N. (1993) *Statistics for Analytical Chemistry*, 2nd edn, Prentice-Hall, Hemel Hempstead.
11 Wallace, R.M. and Katz, S.M. (1964) A method for determination of rank in analysis of absorption spectra of multicomponent systems. *J. Phys. Chem.*, **68**, 3890–3892.
12 Katakis, D. (1965) Matrix rank analysis of spectral data. *Anal. Chem.*, **37**, 876–878.
13 Malinowski, E.R. and Howery, D.G. (1980) *Factor Analysis in Chemistry*, John Wiley & Sons, Inc., New York.
14 Lindsay, R.K., Buchanan, B.G., Feigenbaum, E.A. and Lederberg, J. (1980) *Applications of Artificial Intelligence for Organic Chemistry: The DENDRAL Project*, McGraw-Hill, New York.
15 Kowalski, B.R., Jurs, P.C., Isenhour, T.L. and Reilly, C.N. (1969) Computerized learning machines applied to chemical problems: interpretation of infrared spectrometry data. *Anal. Chem.*, **41**, 1945–1949.
16 Kowalski, B.R. (1980) Chemometrics. *Anal. Chem.*, **52**, R112–R122.
17 Kowalski, B.R. (ed.) (1984) *Chemometrics: Mathematics and Statistics in Chemistry*, Reidel, Dordrecht.
18 Sharaf, M.A., Illman, D.L. and Kowalski, B.R. (1986) *Chemometrics*, John Wiley & Sons, Inc., New York.
19 Massart, D.L., Vandeginste, B.G.M., Deming, S.N. *et al.* (1988) *Chemometrics: A Textbook*, Elsevier, Amsterdam.
20 Massart, D.L., Vandeginste, B.G.M., Buydens, L.M.C. *et al.* (1997) *Handbook of Chemometrics and Qualimetrics Part A*, Elsevier, Amsterdam.
21 Vandeginste, B.G.M., Massart, D.L., Buydens, L.M.C. *et al.* (1997) *Handbook of Chemometrics and Qualimetrics Part B*, Elsevier, Amsterdam.
22 Tauler, R., Walczak, B. and Brown, S.D. (eds) (2009) *Comprehensive Chemometrics*, Elsevier, Amsterdam.
23 Otto, M. (2016) *Chemometrics: Statistics and Computer Applications in Analytical Chemistry*, 3rd edn, Wiley-VCH Verlag GmbH, Weinheim.
24 Beebe, K.R., Pell, R.J. and Seasholtz, M.B. (1998) *Chemometrics: A Practical Guide*, John Wiley & Sons, Inc., New York.
25 Kramer, R. (1998) *Chemometrics Techniques for Quantitative Analysis*, Marcel Dekker, New York.